

PROVEIT RESEARCH CONTINUATION

Exact Quotient Kernels for Unknot Recognition

Sharp interval mergers, scalar gauges,
and simultaneous cyclic overlap search

8 October 2026

Research status. This report supplies exact local algorithms, proofs, native implementations, independent correctness comparisons and measured performance evidence. It does not establish a general quasi-polynomial unknot-recognition bound. The strongest unconditional complexity improvements apply to explicitly specified subproblems; the complete recognizer retains its previous general complexity status.

Pinned baseline

VladimirReshetnikov/ProveIt
58ee11a97d5fd7f21647c57eefaf3ecc007931d6
Topology/UnknotRecognition/fast

The accompanying archive contains the maintained source overlay, an integration patch, the full article source, reproducible experiments, raw results, tests, provenance and checksums.

Abstract

Three sources of repeated work remain visible in the maintained unknot recognizer and its geometric research components: periodic interval regions are processed separately, scalar endomorphism solvers expand dependent morphism coefficients, and explicit group simplification searches the same relator collection once for each donor. We develop exact quotients for these three situations. First, a Fine–Wilf argument gives the necessary and sufficient overlap threshold $p + q - \gcd(p, q)$ for replacing two periodic interval relations by their gcd relation. An independent Agol–Hass–Thurston implementation uses this rule, and a native adapter counts components, boundary curves and orientability directly from binary normal coordinates in a validated finite torus-boundary triangulation. Second, typed differential blocks are factored through their actual coefficient spans. Invertible coefficient matrices then eliminate redundant endomorphism unknowns along a forest without identifying distinct chain objects. The complete commutant is reconstructed in canonical coordinates, so bounded candidate selection is preserved. Third, a shared index solves the explicit all-pairs cyclic relator-overlap problem; a bounded incumbent search exploits the elementary donor-length bound before invoking that index. The report distinguishes proof, implementation, observed activation and the missing global representation-size bounds throughout. Classical ingredients are attributed explicitly; no priority claim is made for the underlying orbit-counting, periodicity, suffix-automaton or linear-algebra theorems.

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1 The baseline and the research objective

The requested target is an exact unknot recognizer with running time $n^{O(\log n)}$, where n is the size of an ordinary knot diagram. The repository already contains a substantial exact pipeline: diagram simplification, structural certificates, polynomial and group-theoretic filters, specialized braid routes, and scanning Khovanov backends. It also contains separate research kernels for compressed geometric operations. Our baseline is the Git commit printed on the title page, committed on 8 October 2026 at 19:17:11 UTC. All integration comparisons refer to that snapshot; reports created against earlier snapshots are not silently treated as the incumbent implementation.

The present work concerns three concrete computational tasks. The first is geometric connectivity for a normal surface whose coordinate sum may be exponential in the triangulation size. The second is the exact scalar commutant of a typed differential, which is used to discover direct-sum splittings. The third is maximum guaranteed shortening by a single cyclic relator replacement. Each task has a finite, independently checkable specification. Improvements to them can be verified without assuming the desired global recognition theorem.

Lackenby’s 2026 hierarchy paper gives new incompressibility and unknot algorithms, while explicitly leaving a running-time specification for several operations incomplete. Its Proposition 9.1 bounds repeated steps by $L(g+1)^L$ in terms of hierarchy length and pattern complexity. The refined construction supplies a linear bound on hierarchy length in the initial q -complexity. These are significant structural facts, but neither is by itself a quasi-polynomial bound for the encoded implementation [2]. The 2021 lecture announces the intended accelerated framework [3]. We use these as motivation and as a precise list of remaining obligations, rather than interpreting efficient local arithmetic as a proof of the complete algorithm.

1.1 What the report contributes

1. An exact two-period merger criterion, with a necessity argument and an explicit family showing a linear reduction in the number of orbit algorithm cycles. This is a sharpened application of the classical Fine–Wilf theorem within the AHT algorithm.
2. A native, coordinate-compressed topology query for arbitrary admissible normal vectors in its stated class of finite triangulations. It supports disconnected and one-sided surfaces and does not require an extreme-ray witness.
3. Exact coefficient-span and forest substitutions in the scalar endomorphism equations, including reconstruction of the entire original solution space and preservation of its canonical basis.
4. A simultaneous index and a bounded adaptive policy for explicit cyclic relator overlaps, replacing an all-donor repeated scan with a near-linear local query in the specified computational model.
5. Reproducible comparisons against the pinned code and independent small oracles, together with corrections, negative measurements and a concrete research agenda.

The full general recognition bound remains unproved. In particular, this work does not bound intermediate Khovanov multiplicities, the total length of explicit group presentations, the number of relator moves, the complete encoding of a cut manifold, or the number of hierarchy repairs. Those quantities cannot be omitted from the complexity accounting.

2 Representations and cost parameters

It is useful to distinguish an expanded object from its stored description. The relevant parameters are summarized below. A parameter belonging to one subproblem is never substituted for diagram size without a separate proof.

Parameter	Meaning
n	Number of crossings, together with the ordinary diagram encoding.
t, W	Tetrahedron count and sum of the seven normal coordinates per tetrahedron.
N, k	Number of integer points and number of interval pairings in an orbit query.
b	Bit length $\lceil \log_2(N + 1) \rceil$ of interval endpoints.
s_v	Dimension of one scalar type in a differential component.
V	Number $\sum_v s_v^2$ of original scalar endomorphism variables.
ρ_{uv}	Dimension of the span of coefficient values in differential block D_{vu} .
V_F	Number of remaining matrix variables after the chosen invertible forest.
m, ℓ	Number of nonempty explicit relators and their total expanded letter length.

An interval pairing stores a constant number of b -bit integers. A normal vector stores $7t$ integers; W may be exponentially larger than that input. The new orbit algorithms operate on endpoints, never on a list of N points. Conversely, the new relator index takes explicit words. It does not turn a straight-line grammar of exponentially long words into a polynomial explicit string. The existing compressed-word route remains a different algorithm with its own contracts.

For binary linear algebra we count bit operations conservatively. A Python integer containing a packed coefficient is not a unit-cost scalar of unlimited size. Its XOR, shift and bit-count costs depend on its length. For the string index, dictionary-operation bounds and bit-complexity bounds are distinguished. These distinctions matter when comparing an apparent constant-size formula with the actual stored input.

3 An exact merger theorem for periodic interval relations

Let $R = [a, z] \cap \mathbb{Z}$ and let $p > 0$. A periodic pairing of period p identifies x and $x + p$ whenever both lie in R . In the implementation it is represented by $[a, z - p] \rightarrow [a + p, z]$. We call it periodic only when its domain width is at least p , equivalently $|R| \geq 2p$. Its orbits in R are exactly the p residue classes modulo p .

Suppose two such relations have supports R_1, R_2 , periods p, q , and overlap of size $h = |R_1 \cap R_2|$. Write $d = \gcd(p, q)$. The original AHT periodic merger uses the sufficient condition $h \geq p + q$ [4]. The following criterion removes the unnecessary d points.

Theorem 3.1 (Exact two-period merger). *For two periodic pairings as above, their generated equivalence relation on $R_1 \cup R_2$ is the residue relation modulo d if and only if*

$$h \geq p + q - d.$$

Thus, under this condition, they may be replaced by a single periodic pairing of period d on the union. The equivalence remains valid in the presence of any additional pairings.

Proof. Every old identification preserves residue modulo d , so it cannot identify points of different d -classes. Color each point of the intersection by its orbit under the two old relations. This finite color sequence has periods p and q : two intersection points separated by either period already lie in one of the periodic supports. If $h \geq p + q - d$, the Fine–Wilf theorem [5] says that the color sequence has period d . Moreover $p + q - d \geq \max(p, q)$, so the intersection contains a representative of every residue class of both original supports. Every point outside the intersection is therefore linked to an intersection point of the same d -class. This proves sufficiency.

For necessity, contract the first support to its p residue classes and the second to its q residue classes. Before identifying their common points there are $p + q$ vertices. Each of the h common points contributes one edge between the corresponding two residue vertices, allowing parallel edges but no further identifications. A graph with $p + q$ vertices and h edges has at least $p + q - h$ connected components. If $h < p + q - d$, this exceeds d . The residue relation modulo d has exactly d classes on the union, so replacement would change the equivalence relation. Finally, equality of the two generated relations is preserved when a common collection of additional generators is adjoined. $\square$

The necessity statement concerns these two generators alone. Other pairings can create extra connections and can justify mergers below this threshold. The implementation never infers that a failed two-generator test proves global disconnectedness.

Example 3.2 (The endpoint matters). The pairings $[0, 3] \rightarrow [2, 5]$ and $[2, 4] \rightarrow [5, 7]$ have periods 2, 3 and an overlap of four points. They generate a connected relation and can be replaced by a period-one relation. The original five-point condition misses this merger. Moving the second support one point to the right leaves only three common points and produces two orbits; the period-one replacement would then be incorrect.

3.1 A family with a linear reduction in orbit cycles

Proposition 3.3 (Adjacent periodic blocks). *For $r \geq 1$ and $p \geq 1$, consider r pairings*

$$[ip, (i+1)p-1] \longrightarrow [(i+1)p, (i+2)p-1], \quad 0 \leq i < r,$$

on $N = (r+1)p$ points. Both orbit algorithms return p . With the stated maximal-transmission and suffix-truncation policy, the original AHT merger requires $r+1$ cycles, while the sharp merger requires two.

Proof. Each pairing joins equal positions in two consecutive blocks. Therefore the p positions within a block label the complete orbits. Neighboring supports overlap in p points, below the old $2p$ threshold, and nonneighbors overlap in fewer points. The rightmost pairing is the only one meeting the last block. No other range is contained in its range, so one suffix truncation deletes this block and this pairing. Induction leaves one block after r cycles; a final static contraction counts its p points.

Under Theorem 3.1, neighboring period- p supports meet the exact threshold p . Repeated merging combines all pairings in the first cycle. The resulting single pairing sends $[0, rp-1]$ to $[p, (r+1)p-1]$. Suffix truncation removes its entire range, leaving $[0, p-1]$; the second cycle counts these p static points. $\square$

This is a cycle-count theorem for an explicit interval family, not a family of knot diagrams. The straightforward implementation additionally spends time scanning candidate pairs. The benchmark records both cycles and candidate tests rather than using cycle count as a proxy for elapsed time.

4 Native orbit counting and signed components

The implementation `fastunknot/interval_orbits.py` follows the classical unweighted AHT reduction. Identity relations are removed, uncovered intervals are counted and contracted, overlapping reflections are trimmed, valid periodic pairs are merged, other ranges are transmitted through a maximal rightmost pairing, and its exclusive suffix is removed. Powers of a translation are evaluated by integer division, avoiding an iteration per translated point. These transformations and the polynomial termination theorem are classical [4]; the sharp merger is the specific change analyzed in Section 3.

4.1 Why the stronger rule retains polynomial termination

The AHT analysis uses a multiplicative quantity of the form $X = 4^k \prod_i w_i$, where w_i are pairing widths. It is not necessary for every individual cycle to halve X . The classical proof bounds the number of cycles between sufficient reductions of this quantity.

For a sharp merger, the two old widths are w_1, w_2 and the new width is

$$w' = w_1 + w_2 + p + q - h - d \leq w_1 + w_2 \leq 2w_1w_2.$$

Thus k decreases by one and the contribution to X drops by at least a factor of two. A state with no sharp merger also has no original AHT merger, so the hypotheses needed by the remaining classical progress argument still hold. The strengthened reduction therefore keeps a polynomial bound in k and $\log(N+1)$. The implementation uses simple lists and pair scans; it is not claimed to attain the smallest possible polynomial exponent.

The stored number of pairings never increases in a count query. All endpoints remain within the original interval. Arithmetic has bounded bit length, and the largest translation powers are handled in one division. The number of represented points is allowed to be vastly larger than memory because it is never used as an iteration bound.

4.2 Signed gluing and orientation consistency

An order-reversing map of a sheet interval need not reverse surface orientation. These are distinct pieces of data. For a pairing g_i , let $\epsilon_i \in \mathbb{F}_2$ be a separately specified parity. A component is consistent when the parity sum around every closed walk is zero.

Proposition 4.1 (Two-sheeted count). *Let C be the number of original orbits and $\hat{C}$ the number of orbits of*

$$(x, e) \sim (g_i(x), e + \epsilon_i), \quad e \in \mathbb{F}_2.$$

Then the numbers of consistent and inconsistent components are

$$C_+ = \hat{C} - C, \quad C_- = 2C - \hat{C}.$$

The lifted system is represented by $2k$ interval pairings on $2N$ points, so both counts have polynomial bit complexity in the original description length.

Proof. Fix a base component and one lift of a chosen point. If all closed walks have parity zero, continuation of this lift determines one sheet at every point; the two choices of initial sheet give two disjoint lifted components. If some closed walk has odd parity, the two initial sheets are connected and the lift is connected. Thus $C = C_+ + C_-$ and $\hat{C} = 2C_+ + C_-$. Solving the two equations gives the formulas. Two consecutive blocks of N integers encode the sheets; each old interval pairing gives one map from each sheet to its prescribed target sheet. $\square$

Proposition 4.2 (Point membership without extraction). *Two supplied points x, y belong to the same orbit if and only if adjoining the singleton relation $x \sim y$ leaves the number of orbits unchanged. If they were in distinct orbits, this addition reduces the number by exactly one.*

This gives an exact membership query using two counts. It does not extract the normal coordinates or attachment data of a chosen component. Weighted AHT, which is a classical extension, remains a separate implementation task.

5 Binary normal-surface topology in a supplied triangulation

The new adapter accepts a nonempty finite tetrahedral face-pairing description of a connected compact orientable 3-manifold with exactly one torus boundary. Its native validator checks reciprocal face gluings, edge orientations, edge links, vertex links, orientability, connectedness and boundary topology. This validator is adapted, with attribution, from the earlier report 36 normal-disc certificate. The current extension accepts every nonnegative admissible normal vector in that ambient class; it does not impose primitivity or an extreme-ray rank condition.

Normal coordinates in each tetrahedron are

$$(T_0, T_1, T_2, T_3, Q_{01|23}, Q_{02|13}, Q_{03|12}).$$

All matching equations and the condition of at most one nonzero quadrilateral type per tetrahedron are checked using exact integers. Hexadecimal integer input is supported. The empty surface is represented by the zero vector and has no components.

5.1 The arc-pairing construction

Give each global triangulation edge a fixed direction. A parity union-find transports local edge directions through face gluings; using vertex labels alone would be wrong in a one-vertex triangulation. If a global edge meets the normal surface w_e times, allocate an interval of w_e points ordered along this direction. The disjoint union of these intervals has

$$N = \sum_e w_e$$

points. Each tetrahedral normal disc meets at most four edges, so $N \leq 4W$.

In a triangular face opposite vertex f , the normal arcs cutting off a vertex v occur with multiplicity

$$a_{fv} = T_v + Q_{fv},$$

where Q_{fv} denotes the quadrilateral type disjoint from the edge $\{f, v\}$. These arcs join the a_{fv} endpoints nearest v on its two incident face edges, in their common order from v . Consequently the whole parallel family is one interval isometry, possibly reversing the globally chosen edge orders. One copy of every glued face and every boundary face suffices. There are at most three pairings per face, and therefore $O(t)$ pairings overall.

Theorem 5.1 (Normal-surface topology query). *For an admissible normal surface S in the supported triangulation class, the adapter and unbounded orbit kernel compute the following in time polynomial in t and $\log(W + 1)$:*

1. the number of components of S ;
2. the numbers of orientable and nonorientable components;
3. the number of boundary curves of S ;
4. the total Euler characteristic of S .

When S is connected these data determine its genus or crosscap number. No normal disc or edge intersection point is expanded.

Proof. The constructed graph is the one-skeleton of the normal cell structure, with each parallel arc family encoded as an interval pairing. Every normal disc is attached along a connected polygonal boundary. Attaching such discs does not merge different components of the one-skeleton. Conversely, every point of a disc can be joined inside the disc to its boundary. Thus graph components and surface components are identical. This is the classical normal-surface reduction to AHT connectivity [4], now made explicit in the native coordinate adapter.

For the boundary count, retain only boundary faces and boundary edge intervals. Every vertex of this normal boundary graph has degree two, so its components are precisely the boundary circles. Interior edge points must be removed from this query; leaving them as isolated vertices would falsely increase the boundary count.

The normal surface with coordinates $2S$ is the horizontal boundary of the normal interval bundle of S . In an orientable ambient 3-manifold this is the orientation double cover. If its component count is C_2 , then $C_2 - C$ original components are orientable and $2C - C_2$ are nonorientable, by the same covering argument as Proposition 4.1. The vertical annuli over ∂S are not included in this statement.

Finally, the cell counts are obtained arithmetically: vertices are $\sum_e w_e$, edges are the sums of normal-arc multiplicities over each face taken once, and faces are the W discs. Hence

$$\chi(S) = \sum_e w_e - \sum_{\text{global faces } F} \#(S \cap F) + W.$$

For a connected orientable surface of genus g with b boundary curves, $2g = 2 - b - \chi(S)$; for a connected nonorientable surface the crosscap number is $2 - b - \chi(S)$. Construction and validation use $O(t)$ combinatorial records and binary arithmetic. The three orbit systems have $O(t)$ pairings and endpoint lengths $O(1 + \log(W + 1))$, completing the complexity claim. $\square$

5.2 Compressing discs and the correspondence still needed

If S is connected, has Euler characteristic one and one boundary curve, it is a disc. On a torus, an embedded simple closed curve is essential exactly when its mod-two homology class is nonzero. The adapter computes the dual boundary cocycle by taking $w_e \bmod 2$ on each boundary edge. It tests whether this cocycle is a vertex coboundary by parity propagation. A nonzero class therefore proves that the disc compresses the supplied boundary torus.

The output is a relative statement about this triangulation. It is not a PD-level unknot certificate: the triangulation-to-knot correspondence is not established by this module. Nor does the module search for the normal vector. A successful topology calculation must not be substituted for a missing exterior-construction certificate.

5.3 Large multiplicities and one-sided surfaces

For a two-sided normal disc D , the vector aD represents a parallel discs. The output component count may therefore be a 500-bit integer while the description uses only a fixed number of coordinates. For a one-sided normal Möbius band M , the vector aM has $\lfloor a/2 \rfloor$ annuli and, when a is odd, one Möbius band. In particular,

$$C(aM) = \lceil a/2 \rceil, \quad C_+(aM) = \lfloor a/2 \rfloor, \quad C_-(aM) = a \bmod 2.$$

These formulas are independent checks of the doubling calculation. The tests include multiplicities exceeding 2^{500} .

The inherited Fibonacci layered-torus family has t tetrahedra and a normal meridian containing $F_{t+5} - 5$ discs. It gives a concrete geometric separation between the coordinate encoding and expansion. The article reports this constructor as inherited, not as a new topological family. The new computation is its general component and orientation analysis.

6 Factoring scalar equations through coefficient spans

Consider one finite differential component in the scanner. The scalar types are the triples of homological degree, boundary matching and recovered quantum shift. If type v occurs s_v times,

an allowed scalar endomorphism is a matrix $Q_v \in M_{s_v}(\mathbb{F}_2)$. Distinct types remain distinct objects. Commutation means

$$Q_v D_{vu} = D_{vu} Q_u$$

for every typed differential block. The coefficients of D_{vu} are packed morphisms in a fixed binary vector space for that ordered pair of types. The original solver expands separate equations for individual nonzero basis coefficients.

Theorem 6.1 (Coefficient-span reduction). *Choose a basis $g_1, \dots, g_\rho$ of the $\mathbb{F}_2$ -linear span of the entries of D_{vu} . Write uniquely*

$$D_{vu} = \sum_{i=1}^{\rho} g_i A_i, \quad A_i \in M_{s_v \times s_u}(\mathbb{F}_2).$$

Then the scalar commutation equation is equivalent to

$$Q_v A_i = A_i Q_u \quad (1 \leq i \leq \rho).$$

In particular, the complete scalar endomorphism space is unchanged.

Proof. Expand the difference as

$$Q_v D_{vu} - D_{vu} Q_u = \sum_i g_i (Q_v A_i - A_i Q_u).$$

For each matrix position this is a linear combination of independent g_i , so it vanishes if and only if each scalar coefficient vanishes. No multiplication of two morphism coefficients is used here; ordinary linear independence suffices. Applying the argument independently to every ordered type pair proves equality of the global solution spaces. $\square$

If e_{vu} is the number of nonzero entries in this block, then

$$\rho_{vu} \leq \min(e_{vu}, \dim \text{Hom}(u, v)).$$

The number of generated matrix-entry equations is at most

$$P_{\text{span}} = \sum_{u,v} \rho_{vu} s_u s_v.$$

Many equations may be zero or dependent, but this bound holds before binary elimination. This coefficient count is optimal among scalar-matrix expansions of the block: if $D_{vu} = \sum_{j=1}^r h_j B_j$, every entry lies in $\text{span}(h_1, \dots, h_r)$, so $r \geq \rho_{vu}$. The span construction attains this lower bound. This is a statement about the number of coefficient factors, not about the fastest way to solve all resulting equations.

In comparison, expansion through a large ambient monomial basis can create many redundant equations even when all entry values lie in a one- or two-dimensional span.

The implementation performs Gaussian reduction on the packed morphism values themselves, recording coordinates in the newly discovered independent residuals. It then constructs each A_i directly as a binary matrix. It does not enumerate all set monomial bits to build the span. A separate reconstruction test verifies that the factorization reproduces every original matrix entry.

6.1 What this does and does not improve asymptotically

Let B bound the packed bit length of the coefficient values in a block. A direct factorization uses at most $e\rho$ reductions of B -bit values, costing $O(e\rho B)$ bit operations up to ordinary indexing factors. The implementation also builds the coordinate matrices as packed target columns: setting a bit in a column can cost $O(s_v)$ bit operations. Including this construction, a conservative

bound is $O(\epsilon\rho(B + s_v) + \rho s_u s_v)$ bit operations up to indexing factors; the final term also accounts for the possible coordinate-matrix output size. Binary elimination on P rows in V variables has the elementary upper bound $O(PV^2 + V^3)$ bit operations. Substituting P_{span} can remove a large dependent-equation factor.

The ambient packed bit length B is still an input cost. This result does not prove polynomial time in frontier width when individual morphisms themselves have exponentially large supports. It also does not reduce the number of chain objects before the scalar splitter is called. Its exact benefit is to replace an ambient coefficient description in a linear constraint system by its observed rank.

7 Eliminating matrix unknowns along an invertible forest

An invertible coefficient matrix carries more information than its individual commutation equations. If $s_u = s_v = s$ and A is invertible, then

$$Q_v A = A Q_u \iff Q_v = A Q_u A^{-1}.$$

This eliminates one entire $s \times s$ unknown. The operation is a substitution in the endomorphism equations. It is not a change of the chain-complex type or an identification of different matchings.

Theorem 7.1 (Forest gauge for the complete commutant). *Form a graph on scalar types using selected invertible coefficient matrices between equal-dimensional types, and choose a forest F . Root each tree. Let P_v be the product of the chosen matrices and their inverses along the root-to- v path, with $P_r = I$ at its root. Then the full original commutant is in bijection with the solutions in root matrices X_r of*

$$X_{r(v)} \tilde{A}_i = \tilde{A}_i X_{r(u)}, \quad \tilde{A}_i = P_v^{-1} A_i P_u,$$

over every original coefficient matrix, through the reconstruction

$$Q_v = P_v X_{r(v)} P_v^{-1}.$$

The reduced number of variables is

$$V_F = \sum_{r \text{ a root}} s_r^2 \leq V.$$

Proof. Each selected forest edge determines its target endomorphism uniquely from its source. Since a tree has a unique path from its root, induction along this path proves the reconstruction formula for every original solution. Substituting that formula into an arbitrary original equation and multiplying by P_v^{-1} on the left and P_u on the right yields precisely the displayed root equation. Conversely, root matrices satisfying all these equations reconstruct matrices satisfying every original equation. The two maps are inverse because each root has $P_r = I$. $\square$

The reconstruction also preserves addition, multiplication and identity, because each component is conjugation by P_v . It is therefore an isomorphism of endomorphism algebras, not only of vector spaces. Idempotents and their orthogonality are preserved under this map.

Selected tree equations transform to identities. Non-tree edges, including cycles, still impose constraints; they must not be discarded. Rectangular blocks between different-dimensional trees also remain as intertwiners. The implementation tests candidate coefficient matrices for invertibility, keeps a forest, and transports all remaining equations. It is not claimed to find every invertible linear combination in a coefficient matrix space.

7.1 Canonical reconstruction preserves bounded search

The maintained splitter does not exhaust the whole endomorphism algebra. It examines a deterministic bounded sequence derived from a canonical nullspace basis, then verifies candidate decompositions against the typed differential and all transformed attachments. Returning merely some basis of the same space could change this sequence and hence change what the bounded search discovers.

After forest elimination, each reduced basis vector is lifted to the original row-major variable order. The lifted span is reduced to the same canonical binary basis used by the original nullspace solver. Uniqueness of reduced echelon form implies equality of the ordered basis, not merely equality of its dimension. Consequently the downstream candidate sequence and accepted witnesses are preserved whenever the original scalar-type and connected-component preconditions hold.

The reconstruction itself costs time. The implementation precomputes the rank-one images of root matrix units when useful, then combines them to lift basis vectors. Complete-solve timings include this step. A reduction in V is not reported as a runtime improvement unless the transport and canonicalization costs are also included.

7.2 Interactions with previous commutant reuse

The previous optional child-commutant reuse restricts a complete parent algebra after a verified direct-sum split. Coefficient-span and forest reduction accelerate a fresh solve; they do not justify using a stale algebra after a differential change. These operations can be combined within a static compression pass, but a scan step, cancellation or other differential mutation requires the existing invalidation discipline.

The original object and variable caps are applied before the new solve. Thus a small V_F does not currently allow an otherwise oversized component to bypass the caller's cap. This conservative resource contract simplifies integration, and is explicitly separated from a possible future reduced-variable admission policy.

8 A shared index for maximum cyclic relator shortening

8.1 The bottleneck and the exact query

The existing native positive-certificate path reconstructs the complete Wirtinger presentation, performs generator eliminations and Whitehead moves, and searches for a length-reducing relator substitution when these moves stall. The relevant local query takes an indexed list $(R_1, \dots, R_s)$ of explicit cyclic words. Let m be the number of nonempty slots and $L = \sum_i |R_i|$. For distinct slots $i \neq j$, rotate R_i and either R_j or R_j^{-1} to write

$$R_i \sim UW, \quad R_j^\varepsilon \sim UV, \quad \varepsilon \in \{1, -1\}.$$

Replacing the target by $V^{-1}W$ while retaining the donor preserves the normal closure. Indeed, $(UV)^{-1}(UW) = V^{-1}W$ and $(UV)(V^{-1}W) = UW$. The guaranteed decrease before incidental free cancellation is

$$g = 2|U| - |R_j|.$$

The search must find the maximum positive g over all target rotations, all donor rotations and both donor signs, or prove that none exists. No small-cancellation hypothesis is needed for the soundness of a move. Stalling is inconclusive for unknot recognition.

The maintained pairwise implementation builds a suffix automaton for each oriented doubled donor and scans every eligible doubled target. Its bound is $O(s + mL)$ dictionary operations and $O(L)$ auxiliary space. Profiling the actual 141-crossing Gordian input found one such query after 129 generator eliminations. Twelve nonempty relators then have total length 17 504. This query,

rather than the elimination loop, was the dominant producer cost. The new query removes the factor m from its asymptotic operation bound; it does not bound the number of subsequent presentation moves.

8.2 Cyclic windows as capped points on a suffix-link tree

A nonempty word w of length d has all its cyclic rotations among the length- d windows ending at positions

$$d-1, d, \dots, 2d-2$$

of $ww[0 : d-1]$, using zero-based positions. There are exactly d such windows, even when several spell the same word. Each shorter cyclic segment is a suffix of one of them. A capacity d prevents any accepted match from wrapping around the donor or target twice.

Build one suffix automaton for the concatenation of the positive doubled relators, with a separator after each block. In the implementation the separator is a fresh object outside the signed-generator alphabet. The indexed string has length exactly $2L$. Standard suffix-automaton theory provides $O(L)$ states and transitions; the classical linear-size subword index is due to Blumer and coauthors [7]. A state has a maximum represented substring length $\lambda(v)$ and a suffix link $p(v)$ satisfying $\lambda(p(v)) < \lambda(v)$. The represented lengths on its suffix-link edge are the integers

$$\lambda(p(v)) + 1, \dots, \lambda(v).$$

All strings represented by one state are suffixes of its unique longest string. Consequently a virtual point at an intermediate edge depth has a well-defined word, not merely a numerical length. Thus an occurrence of a word of length d can be represented by a point at depth d on the suffix-link path of its ending state. Insert that point as a terminal cap if it is not already a state vertex.

For every positive rotation, mark its capped point with two roles: it is a possible target and a possible positive donor. The mark retains its original relator slot, its word length, its sign, and the ending position inside the doubled source. Equal words in distinct slots retain distinct identities. Losing this information would permit an invalid self-donor substitution.

The inverse donors need not enlarge the index. Scan each inverse doubled relator through the already completed automaton, maintaining the longest matched suffix. At each of its d cyclic-window endpoints, let h be the length of that matched suffix. If $h > 0$, attach an inverse-donor mark at depth $\min(d, h)$, but retain the original donor length d for the gain calculation. This is a partially represented inverse window, not a claim that its full length- d rotation occurs in the indexed text.

Lemma 8.1 (Completeness of the inverse-window marks). *Every common cyclic segment of an inverse donor and a positive target is represented by the common ancestral path of one inverse-donor mark and one positive-target mark. Conversely, every such common path denotes a literal common segment of those oriented relators.*

Proof. Let the segment have length u . Choose the donor's length- d cyclic window ending with this segment; its endpoint is one of the d recorded endpoints. Since the segment occurs in a positive target, it occurs in the indexed text. The longest suffix match at that donor endpoint has length at least u , and $u \leq d$, so the inverse mark has depth at least u . Its suffix-link path therefore represents the required segment. The positive target has an analogous complete cyclic-window mark. Conversely, an ancestor of a mark represents a suffix of the actual marked window. The depth caps ensure that this suffix has length at most the corresponding original relator length. A common ancestor represents the same substring in both windows. $\square$

The capped tree contains at most $2L$ marked windows and at most $2L$ new cap vertices. Together with the automaton states it has $O(L)$ vertices. Marks can coincide, in which case their original slots remain separate records. No quadratic list of explicit rotations is constructed.

8.3 Two representatives suffice for the distinct-slot constraint

At a capped-tree vertex of depth h , all descendant marks contain the same length- h suffix. A donor of length d and any distinct-slot target therefore supply gain $2h - d$. At this fixed depth, maximizing gain means minimizing donor length among eligible donor slots.

Lemma 8.2 (Two-slot summary). *At each tree vertex, retain the two least donor records of distinct original slots, ordered first by donor length, and any two target records of distinct slots. These summaries suffice to recover the largest guaranteed gain of a distinct-slot donor–target pair available at that vertex.*

Proof. Consider the shortest retained donor slot j . If there is a target of a different slot, one of two distinct target representatives supplies it: at most one can have slot j . This donor is optimal. If no such target exists, every target has slot j . The shortest donor with a different slot is then the second retained donor; if it does not exist, no eligible pair exists. The same argument covers vertices with only one target or one donor slot. Donor orientation and occurrence location do not affect the gain at this fixed depth, so a single representative per donor slot is sufficient. $\square$

These constant-size summaries merge bottom-up. The implementation keeps deterministic tuple orders for witnesses, but does not promise the historical suffix automaton’s tie choice. Maximum gain, distinct slots, valid offsets and literal equality are the required invariant properties.

Theorem 8.3 (Exact shared-index query). *The shared capped suffix-link index returns a maximum positive guaranteed decrease over all distinct relator slots, cyclic rotations and donor signs. It uses $O(s + L \log(L + 1))$ integer and dictionary operations and $O(L)$ auxiliary storage. Under the usual expected constant-time hash-dictionary assumption this is the corresponding expected word-time bound.*

Proof. Every candidate common cyclic segment is represented by two marked points, by construction and the inverse-window lemma. Their longest common suffix ends at their lowest common ancestor in the tree after cap insertion. The gain is increasing in the common depth when the donor is fixed, so an optimum is attained at a tree or cap vertex. The two-slot lemma recovers an optimal pair at every such vertex. Taking the largest of these values therefore gives the global optimum, and each recovered pair is a literal valid match.

The indexed text has length $2L$. Building its suffix automaton and scanning all inverse blocks uses $O(L)$ dictionary operations. A depth-first traversal maintains the increasing depths on its current suffix-link path; binary search locates each of at most $2L$ terminal caps in $O(\log(L + 1))$ operations. Caps on each original edge are sorted. Their total sorting cost is $O(L \log(L + 1))$, since the sum of their counts is $O(L)$. Constant-size summary propagation and candidate evaluation are linear in the capped tree size. The original list of s slots is traversed once. All stored arrays, transitions, window marks and cap records have linear total size. $\square$

For a node of depth h , with donor endpoint e_d and target endpoint e_t , the certificate offsets are

$$a_d = (e_d - h + 1) \bmod d, \quad a_t = (e_t - h + 1) \bmod t.$$

The resulting record uses the existing version-two relator fields. No change to the certificate format or the independent checker is needed.

8.4 A bounded prelude that improves the actual knot query

The shared index alone is slower on Gordian than the small set of useful pairwise scans. This measured regression motivates the implemented adaptive query. As a fixed-size base case, lists with at most four nonempty relators use the existing pairwise query. Its cost is $O(s + L)$ in that case, and its historical tie choices are preserved. The threshold four is fixed independently of

the input; allowing it to grow with the input would require including it as a parameter in the complexity bound. This base case avoids paying for both a failed prelude and a shared index on the two small mirror examples. For larger lists, every donor of length d satisfies

$$2 \min(d, t) - d \leq d.$$

Consequently a donor with $d \leq G$, where G is the largest gain already found, cannot improve the incumbent. Process donors in decreasing length, retain the existing target upper bound, and stop once all remaining donor lengths are at most G .

This is a sound search order and pruning rule, but on its own retains the old worst-case mL scan cost. The implementation therefore limits the prelude to

$$Q = 2 \max(1, L) \text{ bitlength}(L + 1)$$

charged work units. Sorting, donor construction, suffix-link traversal, transition creation, clone copying, target-pair consideration and target scanning consume this allowance. When the local allowance is exhausted, the complete shared index is built under the same enclosing resource budget. Genuine global cancellation or resource exhaustion propagates; it is not converted into an invitation to reset the budget.

Corollary 8.4 (Adaptive exact query). *The adaptive query has the same maximum-gain guarantee and $O(s + L \log(L + 1))$ dictionary/word-operation bound as the shared query. It retains $O(L)$ auxiliary storage, including failed prelude construction.*

Proof. The fixed-slot base case has linear cost and is exact. Otherwise, if the prelude completes, all unvisited donors and targets have valid upper bounds no larger than the incumbent. It is exact. Otherwise the complete shared query is exact independently of the partial prelude. Prelude work is bounded by Q plus one already charged operation batch of $O(L)$; the shared query then has the preceding theorem’s bound. The one-donor index is no longer needed when the continuation begins. No list of all donor automata or pairwise matches is retained. $\square$

Dictionary-operation bounds should not be confused with unconditional bounds for CPython’s deterministic integer hashing. A deterministic balanced-map implementation gives $O(s + L \log(L + 1))$ comparison operations for the shared index alone and a conservative $O(s + L \log^2(L + 1))$ comparison bound for this adaptive scheduling policy. Comparisons and stored lengths have their ordinary bit costs. The presented Python code uses its existing dictionary conventions; it does not implement a new balanced-map data structure.

For an explicit bit bound on that deterministic variant, let $\beta = \lceil \log_2(L + 2) \rceil$ and let $b_{\text{id}} \geq 1$ bound the bit length of signed-generator identifiers and slot indices. Elementary comparison and arithmetic give the conservative adaptive bound

$$O\left((s + 1 + L\beta^2)(b_{\text{id}} + \beta)\right).$$

The shared index alone has one power of β in place of β^2 . Copied automaton transitions are charged individually, so clone creation introduces no further logarithmic factor. Offset normalization acts on a value in $[0, 2d)$ and needs at most one subtraction. These bounds exclude arbitrary caller-supplied callback work and assume resource counters have comparable bit lengths. They are bounds for the stated balanced-map variant, not a claim that CPython hashing has deterministic constant cost.

On Gordian the maximum guaranteed decrease is $2 \cdot 3362 - 3726 = 2998$. Only the length-4260 and length-3726 donors survive the decreasing-length prelude’s final bound. Four oriented donor automata suffice, versus 24 in the recorded incumbent query. The selected move and the entire 141-move certificate remain identical on this input. The timing table and raw data report both the resulting full-query speedup and the shared-index-only regression.

8.5 Scope and remaining research obligations

This result accelerates an exact query on an explicit presentation. It does not establish a small explicit presentation for every input knot, a bound on grammar growth, a bound on the number of changing relator moves, or completeness of greedy presentation simplification. If L becomes exponential in the original crossing number, even the new local query may take exponential time. The existing compressed-word pipeline continues to handle inputs beyond its explicit expansion cap; this shared explicit index is used only when an explicit query is actually made. No false negative is inferred from an absent shortening move.

For a more explicit conditional accounting, suppose the ordinary explicit producer starts with r generators and its total reduced relator length never exceeds B . There are at most $r - 1$ generator eliminations. Between two successive eliminations, every accepted Whitehead or overlap move strictly decreases an integer length in $[0, B]$, so there are at most B such moves. Including an unsuccessful final query, the number of overlap queries is at most $r(B + 1)$. Their total new cost is therefore

$$O(r(B + 1)(s + B \log(B + 1)))$$

dictionary/word operations. This charges only the overlap searches, not Whitehead cuts, generator substitution, replay or diagram preparation. If B were universally quasi-polynomially bounded, this contribution would be quasi-polynomial as well. Establishing such a bound and ensuring successful recognition remain separate obligations.

A useful next problem is a similarly shared fully compressed cyclic index: can one remove the factor equal to the number of relators while charging only grammar size and logarithmic represented lengths? Another is an incremental index under a certified relator replacement, with deleted occurrences handled exactly. Both require new representation bounds; copying this explicit tree into a compressed arena would not supply them.

9 Experiments, independent checks and negative results

The accompanying data distinguish four scopes: an interval-count query, a topology query on a supplied normal surface, a scalar solve or complete compression pass, and a complete configured knot-recognition query. A speedup at one scope is not promoted to a speedup at another. In particular, a constructed algebraic differential and a synthetic relator list are not classified as hard classical knots.

9.1 Protocol and reproducibility

All reported timings were collected in the same Linux x86-64 environment with CPython 3.12.14. They are elapsed wall times, not universal machine constants. The paired coefficient and whole-query experiments randomize arm order within each round and include a second invocation of the direct algorithm as an A/A control. Input construction and module imports are excluded consistently. Timed complete recognition calls do include fresh PD validation, the configured recognition stages and independent replay. They exclude process startup and writing the result to disk. The raw files retain individual timings, inputs or reproducible seeds, exact validation results or identity hashes, counters, and the relevant source metadata. Coefficient and relator records include explicit arm orders; the normal series stores per-arm samples and its shuffle seed. No conclusion relies only on a rounded summary table.

For paired experiments the reported speedup is the median of the within-round ratios $T_{\text{direct}}/T_{\text{new}}$. It need not equal the ratio of the separately reported medians. Values below one indicate a regression. A/A ratios expose the scale of run-to-run variation; they do not turn this small corpus into a statistical model of all knots. These are deliberately modest repeat counts, with no claim of population-level significance or reliable extrapolation from a fitted exponent.

The coefficient experiments use seven paired rounds with two fresh invocations per arm, except the dense-coefficient experiments, which use five rounds and one invocation per arm. Each family receives an excluded warmup. The relator experiments use five paired whole-query rounds and three kernel rounds, with one excluded warmup per case, mode and arm. Interval timings use three repetitions; their literal union-find controls use one invocation and have lower evidential weight than the exact cycle counts. The normal-surface series uses three shuffled rounds and an excluded warmup, including an A/A arm. A normal expansion control is attempted only below a declared one-million-point limit. Exceeding this limit is neither a timeout nor an observed expansion runtime.

9.2 Periodic interval families

Table 1 instantiates Proposition 3.3 with $p = 2^{512} + 1$. The input consists of k pairings on $(k + 1)p$ points, and the correct answer is p . Its description is small even though the represented point set cannot be materialized.

k	Old cycles	Sharp cycles	Old ms	Sharp ms	Ratio
8	9	2	0.252	0.058	4.3
16	17	2	1.349	0.102	13.2
32	33	2	8.707	0.230	37.9
64	65	2	67.515	0.436	154.8
128	129	2	462.508	0.762	607.3

Table 1: Adjacent periodic interval family. Times are medians in milliseconds. This ratio uses the two medians; it is not a paired statistic.

At $k = 128$ the original rule makes 349 504 merger-candidate tests and 129 cycles. The sharp rule makes 127 tests and two cycles. The elapsed ratio is about 607 in this particular list-based implementation. This large number reflects both earlier merger and avoided repeated pair scans. The proved claim is the exact replacement relation and cycle formula, not an asymptotic speedup factor of 607.

Six independently generated small parity graphs also produce symbolic scaled families with endpoint lengths through 16 384 bits. Their expected counts come from the tiny signed quotient: an orientation-consistent quotient component contributes one orbit per within-block position; an inconsistent component identifies opposite positions. This gives an exact oracle without expanding the large interval. Every reported count agrees. The literal comparison at 1 100 000 points takes approximately 0.344 seconds, versus 0.000064 seconds for the encoded count. That control illustrates the representation gap; it does not compare two optimized implementations of the same compressed algorithm.

9.3 Native topology of large normal surfaces

Table 2 reports the inherited layered-solid-torus meridians. Each timed call validates the triangulation and coordinates, counts components of S and $2S$, counts boundary curves, computes Euler characteristic and tests boundary essentiality. Every row returns one orientable compressing disc. The represented disc count grows according to $F_{t+5} - 5$, while the largest coordinate remains stored in binary.

At 128 tetrahedra the surface has 2 791 715 456 571 051 233 611 642 548 normal discs. The query uses only 514 pairings for its initial surface graph and 397 cycles summed over its three orbit computations. This is a concrete test of encoded topology, including boundary and orientability. It is not a search for a normal meridian and does not establish correspondence with an input knot diagram.

t	Normal discs W	Max. coordinate bits	Total cycles	Time ms
4	29	3	25	1.260
8	228	6	37	3.048
12	1 592	8	49	6.924
16	10 941	11	61	11.581
20	75 020	14	73	18.210
32	$2.4158 \cdot 10^7$	22	109	49.619
64	$1.1767 \cdot 10^{14}$	44	205	318.152
128	$2.7917 \cdot 10^{27}$	89	397	2027.548

Table 2: Layered meridian topology, with no expanded surface construction. Large disc counts are rounded here; exact integers are stored in the JSON.

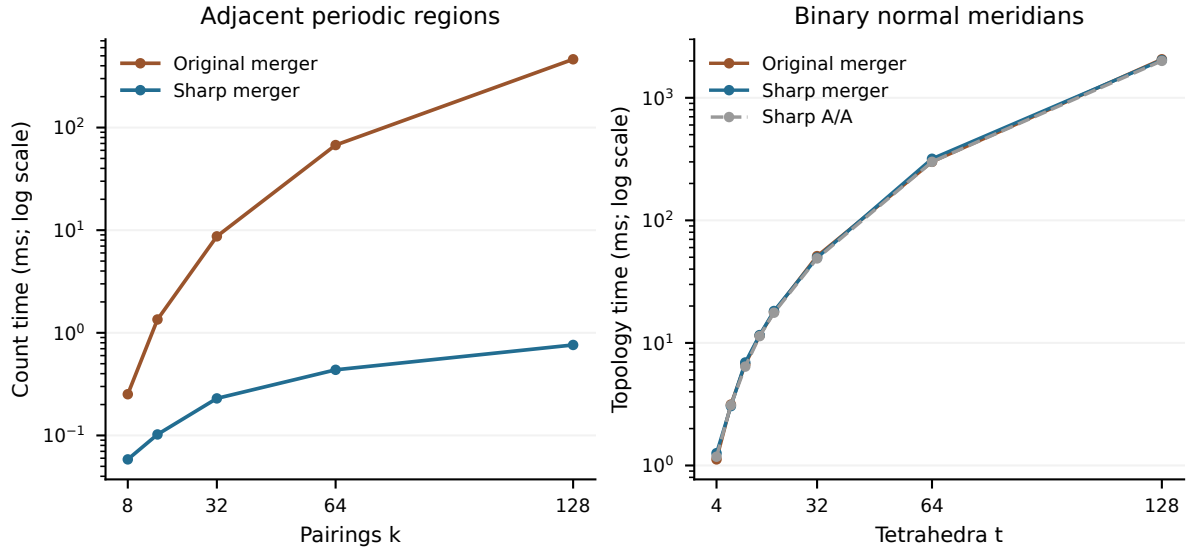


Figure 1: Measured local queries. Left: the adjacent periodic family with $p = 2^{512} + 1$; points join the measured medians and do not represent a fitted asymptotic law. Right: the complete native topology query for the layered meridians, including validation and all three orbit counts. Axes use logarithmic elapsed time. The normal-family curves do not show a uniform timing advantage for the sharper merger.

The general AHT adapter should not replace the inherited specialized extreme-ray disc check on the assumption that it will always be faster. That check can be substantially cheaper on a primitive extreme-ray meridian. The new adapter answers a broader question: it accepts arbitrary admissible normal vectors and handles disconnected and one-sided surfaces. The benchmark also scales a fixed meridian by powers of two through 2^{2000} and verifies the resulting component count exactly. These large-scale checks are supplementary correctness cases, not a timed benchmark with repeated samples.

9.4 Scalar equations on constructed and natural inputs

Two kinds of valid homogeneous two-term algebraic complexes expose the coefficient reduction. The sparse family uses two morphism coefficients and field blocks of degree d . At $d = 10$ it has 40 objects and 800 original scalar variables, within the existing default admission cap. Forest substitution leaves 400 variables. All arms recover the same 20-dimensional full commutant and the same complete compressed complex. The experiment includes the cost of lifting and canonicalizing the basis.

d	Objects	Direct ms	Span ms	Forest ms	Direct/forest	A/A
4	16	1.232	1.217	1.250	0.985	1.012
8	32	7.423	6.051	5.553	1.305	0.957
10	40	12.625	10.207	9.660	1.317	1.000

Table 3: Sparse two-field algebraic family: complete compression-pass medians.

The dense family keeps 16 objects and 128 original variables fixed. Its two independent homogeneous coefficients partition the degree $\lfloor b/2 \rfloor$ dot monomials according to a chosen variable. The actual coefficient rank is two, although an ambient coefficientwise expansion generates thousands of equations. At $b = 12$, the equation count drops from 59 136 to 128 under span reduction, and to 64 after the forest substitution. The forest has 64 remaining variables. Table 4 includes the entire compression pass, not merely Gaussian elimination.

b	Direct equations	Span equations	Direct ms	Span ms	Direct/span	A/A
8	4480	128	12.856	2.000	6.530	0.994
10	16128	128	60.286	4.111	14.666	0.972
12	59136	128	516.973	12.815	38.521	0.996

Table 4: Dense homogeneous coefficients: equations and complete compression timings. Every row has 128 original variables; the forest has 64 variables and 64 equations.

These are mathematically valid chain-complex examples, checked for $D^2 = 0$ and grading consistency. They demonstrate an avoidable ambient coefficient factor. They do not demonstrate that natural knot prefixes attain this gap, nor polynomial time in b : the packed input itself can have length exponential in the number of dot variables.

The natural-knot experiment freezes real scalar-solve inputs from three scans. Summed across their eligible prefix solves, it finds the reductions in Table 5. “Remaining variables” refers to the reduced equation system; every returned basis is still expressed in the full original variable space.

Activation is therefore real. Nevertheless, the complete raw homology scans in Table 6 show no consistent speedup. These scans run the selected Khovanov backend directly, bypassing earlier recognition filters so that they actually exercise its scalar machinery. They are not full recognition-pipeline timings. No ordering search is included in the timed calls; all arms use the same precomputed order, and their graded homology agrees with the ordinary exact reference scan.

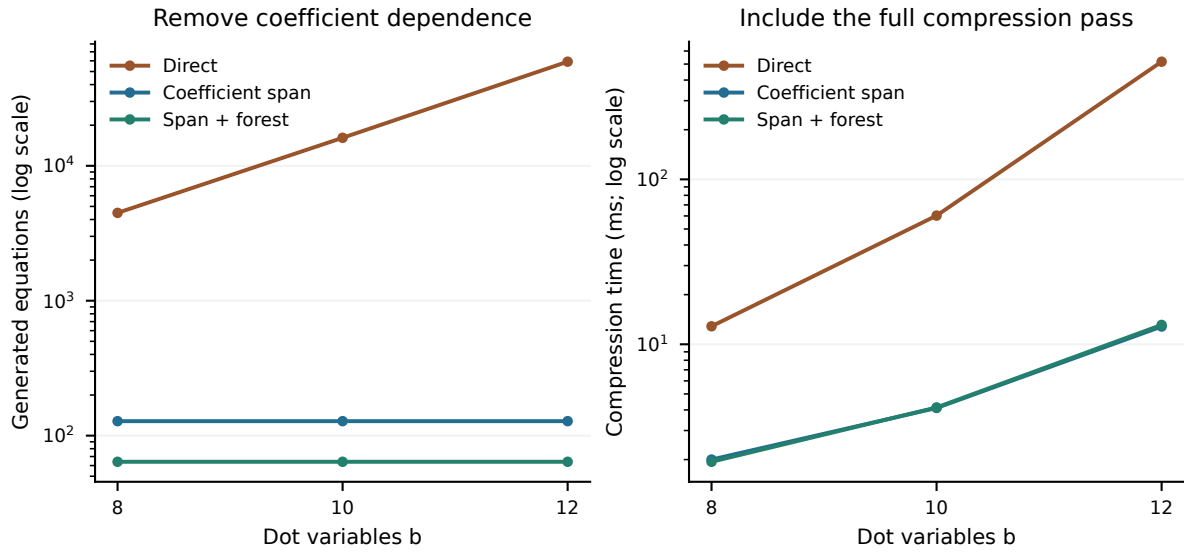


Figure 2: Constructed homogeneous algebraic family, with no claimed classical-knot realization. Left: generated scalar equation counts. Right: complete compression-pass medians, including basis reconstruction. The ambient coefficient values are still packed binary integers; their bit length is part of the input and is not treated as constant.

Input	Solves	Direct eq.	Span eq.	Forest eq.	V	V_F
Conway	10	1310	853	643	590	403
Kinoshita–Terasaka	9	1324	905	699	551	362
5-braid stress (36)	5	237	194	93	97	35

Table 5: Actual prefix commutants: sums of per-solve counts, not one simultaneous system.

Input	Direct ms	Span ms	Forest ms	D/span	D/forest	A/A
Conway	7.917	8.294	8.864	0.959	0.851	0.970
Kinoshita–Terasaka	8.606	9.177	9.836	0.938	0.887	1.024
Hard unknot (8)	0.713	0.757	0.715	1.017	0.949	0.925
5-braid stress (36)	453.776	454.439	451.814	1.000	1.055	1.059
$T(3, 5)$	3.062	3.000	2.865	1.026	1.037	1.044

Table 6: Complete raw homology scans with identical scan order and graded answer. Ratios are median paired ratios. Values near one should be read alongside the A/A control.

Transport overhead, small original systems and work elsewhere in the scan can absorb the reduced solve cost. The maintained default consequently remains **direct**; **span** and **forest** are explicit options. A blanket recommendation to enable forest reduction on every prefix would go beyond these measurements.

9.5 Whole knot queries and the explicit overlap kernel

The relator corpus contains twelve maintained survivor unknot diagrams, two of their mirrors, the 141-crossing Gordian unknot, and four nontrivial controls. Each is run with the explicit and the compressed group-search configuration. Four arms compare the historical pairwise query, its A/A copy, the new adaptive query and the shared index alone. All 760 measured whole queries, together with 152 excluded warmups, complete with the expected result. No censored or inconclusive result is silently converted to a successful timing. Full PD encodings, configurations, status records and certificate hashes are included in the raw JSON.

Search mode	Pairwise ms	Adaptive ms	Joint ms	P/adaptive	P/joint	A/A
explicit	830.706	490.701	931.036	1.766	0.920	1.068
compressed	1357.319	1045.271	1487.001	1.297	0.911	1.008

Table 7: Complete configured Gordian recognition queries, including independent replay. The explicit-query replacement can also be reached inside the configured compressed route.

The Gordian improvement is attributable to one residual overlap query, after 129 generator eliminations. It has twelve nonempty relators of total length 17 504. The maximum guaranteed gain is 2 998, and the adaptive prelude needs four oriented donor automata in place of 24. The selected move and the complete 141-move certificate are unchanged on this input. The independent replay modules do not use the producer’s shared index. Tests also disable producer-only helpers during replay, so accepting the optimized search output is not circular.

The shared index by itself regresses on Gordian. This negative arm is retained in the table. Its larger construction cost is justified for many relators, whereas Gordian admits strong incumbent bounds early. The fixed four-relator base case avoids paying a failed prelude followed by shared construction on very small lists. The initial measurement run, which motivated that base case, is retained separately as calibration data together with the matching index source; it is not mixed into the final medians.

Input	ℓ	Gain	Pairwise ms	Adaptive ms	Joint ms	P/adaptive
Gordian residual	17504	2998	265.052	62.032	416.868	4.232
8×64 random	512	0	8.249	9.809	7.025	0.859
32×64 random	2048	0	129.521	46.800	26.003	2.768
128×64 random	8192	0	2066.899	231.601	185.595	8.798
Periodic duplicates	32768	1024	74.049	4.485	311.439	16.510

Table 8: Exact explicit overlap queries, including all index construction and any failed prelude. Only the first row comes from the measured knot presentation; the others are synthetic capacity cases.

The random relator rows are exact-query capacity experiments, not knot recognition results. Their optimum gain is zero, so a correct query must rule out every eligible shortening. On the 128-by-64 case the simultaneous index removes the repeated target scans, and the adaptive route retains most of the benefit even after charging its failed prelude. The eight-by-64 case is a counterexample to universal practical improvement: the adaptive query can be slower. Highly repetitive duplicate relators favor early donor bounds, again making the adaptive route preferable to always building the shared index. No single timing arm is best on every row.

9.6 Independent correctness and validation coverage

The mathematical specifications are checked by several different oracles. The interval suite compares both merger policies with literal disjoint-set connectivity on 9 882 small two-pairing systems, exhaustive tiny triples and 2 000 randomized systems. It exhausts the relevant two-period thresholds for periods through twelve, checks invalid endpoints and shared budgets, and includes 700 independently counted signed systems. Large integer tests ensure that bit length, rather than represented interval size, determines storage.

The normal-surface suite checks explicit arc graphs, reoriented and relabelled triangulations, the zero surface, vertex-linking discs, essential meridians and one-sided Möbius families. It compares normal surfaces and compatible sums in small layered triangulations against Regina 7.4 for connected components, orientability, boundary count, Euler characteristic and compressing-disc status [10]. It also compares the native ambient validator against Regina on the enumerated one-face gluing cases. Regina is an optional independent test dependency; neither new production module imports it.

The scalar suite reconstructs coefficient entries, exhausts 256 small two-coefficient matrix systems, checks randomized rectangular blocks and compares the canonical complete nullspace in all modes. It verifies actual split witnesses, static child-commutant reuse and the existing admission caps. These tests establish equality of the object used by the bounded candidate policy, not just equality of nullity.

The overlap suite checks 7 056 exhaustive small ordered word pairs, 350 small multisets against a literal all-rotation oracle, and 600 larger multisets against the historical query. Cases include inverse donors, periodic words, zero-length slots, equal words in distinct slots, cap boundaries, cancellation, malformed witnesses and tampered certificates. The exhaustive and differential comparisons require the same optimal gain and a valid attaining witness; the general shared index is allowed to choose a different tied witness. Complete corpus certificates are also compared explicitly.

Full integration gate. With the pinned sibling reference dependencies restored, the untouched baseline runs 768 test methods: 767 pass and one fails with the inherited delayed-input subprocess timeout. The maintained overlay passes all **807 test methods**, with no failures, errors or skips. The integrated run includes the repaired worker regression and 39 additional test methods; many methods contain large exhaustive or randomized families. The baseline and maintained runs report 112.659 and 120.692 seconds respectively. These different-size suite durations are validation records, not a performance comparison. Source hashes are unchanged throughout both runs, and the 446-file baseline staging archive remains byte-identical. Full logs and a machine-readable summary accompany the article.

10 Integration contracts and implementation decisions

The archive supplies an overlay at the original repository-relative paths and a binary-capable Git patch against the pinned baseline. The patch is verified by applying it to a clean reconstruction of the affected baseline files and comparing the resulting bytes with the overlay. It includes the article and reproducibility artifacts; no remote branch or publication is created as part of this delivery.

10.1 Production and research interfaces

The exact interval kernel is a standalone native Python module. A minimal query, run with `fast/` on the import path, is:

```

from fastunknot.interval_orbits import IntervalPairing, count_orbits

p = 2**512 + 1
pairs = [IntervalPairing(i*p, (i+1)*p-1,
                        (i+1)*p, (i+2)*p-1)
         for i in range(128)]
answer = count_orbits(129*p, pairs, max_cycles=1000)
assert answer.complete and answer.orbits == p

```

Endpoints are inclusive and zero-based. A pairing must have equal domain and range widths. `periodic_rule="aht"` selects the original merger for comparison; the default is `"fine_wilf"`. Signed counting and point-membership queries reuse the same exact kernel. A cycle limit is shared across the constituent counts. An exhausted query returns an incomplete result, and a caller-provided cancellation callback propagates its exception. Neither condition is interpreted as a negative theorem.

The normal-surface adapter can be exercised on a supplied layered meridian:

```

from normal_orbit_research.fixtures import layered_torus
from fastunknot.normal_surface_orbits import normal_surface_topology

triangulation, coordinates = layered_torus(32)
topology = normal_surface_topology(
    triangulation, coordinates, max_cycles=10000)
assert topology['status'] == 'COMPLETE'
assert topology['components'] == 1
assert topology['compressing_disk']

```

This result concerns the provided triangulation. It is not a top-level recognition verdict. The inherited constructor is included for research and tests, with provenance pointing to report 36; the production adapter does not depend on that constructor.

The scalar options are deliberately explicit:

```

from fastunknot.scalar_split import fitting_khovanov_rank

answer = fitting_khovanov_rank(
    pd, fitting_primary=True, fitting_coefficient_mode='forest')
# Other modes: 'direct' (the maintained default), 'span'.

```

The lower-level scalar-space and scalar-split functions likewise accept `coefficient_mode`. The full original basis is returned in every mode. New counters expose coefficient rank, equation counts, forest edges, original variables and reduced variables. Existing admission caps, type distinctions, witness checks and cache-invalidation rules are preserved.

The explicit overlap entry point now defaults to `backend="adaptive"`. The optional values `"pairwise"` and `"joint"` isolate the incumbent and shared-index strategies for audits. This change applies when an existing configured group search calls the explicit overlap query; it does not automatically enable an otherwise disabled group stage or remove an explicit expansion cap. The existing certificate fields and independent replayers remain unchanged.

10.2 Two correctness repairs exposed by the audit

The public scalar-space shortcut for an all-singleton closed group formerly returned only the common identity. For a disconnected declared group, the correct space contains one component-indicator scalar for each weak component. The empty group similarly has a zero-dimensional endomorphism space, rather than a basis list containing the zero vector. The shortcut now handles these cases and rejects malformed duplicate or nonclosed groups. Internal connected scanner calls retain their prior mathematical behavior. These statements concern the declared scalar partition; the change does not recover unknown relative quantum shifts between unrelated components.

The maintained normal-surface subprocess wrapper also had a reproducible partial-input deadlock in this CPython environment. Its polling loop sent the payload in the first `communicate` call and retried with no input after a timeout. If the child delayed reading a large payload, later polls could leave the unsent remainder pending indefinitely. The existing regression test exposed this failure in the untouched baseline.

The repair stages UTF-8 input and captures output in private temporary standard streams, then polls the child’s public `wait` interface. This avoids repeated partial pipe communication. Cancellation and deadline checks cover preparation, waiting and output collection; exceptional exits kill and reap the child and close the streams. The tests retain strict cleanup checks and add Unicode plus large-output coverage. This is a correctness repair, with a temporary-storage tradeoff, and is not included in the reported quotient-kernel speedups.

10.3 Claims at the integration boundary

Component	Established here	Remaining boundary
Interval orbits	Exact counts; sharper merger; polynomial encoded query	Weighted component data and attachment extraction
Normal topology	Native validation and topology for an admissible supplied vector	Surface search, cut output and knot-exterior correspondence
Scalar solve	Same full canonical commutant with fewer equations or unknowns	Uniform bounds on multiplicities, coefficient encoding and reconstruction
Cyclic overlap	Exact maximum guaranteed shortening with a near-linear local query	Total presentation growth, successful move count and completeness
Complete pipeline	Passing integration tests and measured Gordian improvement	A general quasi-polynomial worst-case proof

The integration favors explicit contracts: each reusable local operation states its input representation, its resource behavior, and what its answer certifies. This makes later research failures informative. For example, if a compressed cutting implementation needs information absent from an orbit-count result, it must extend that result’s contract rather than quietly reconstruct an exponentially large surface.

11 Sufficient global bounds and a concrete research agenda

The local theorems expose several quantities that could support a global bound. At present these quantities describe an execution; they are not proved restrictions on every diagram. The next task is to convert selected observations into uniform bounds, while retaining an algorithm that always finishes with a correct decision.

11.1 A sufficient accounting principle

Let $S_*(n)$ bound the complete bit length of every state, query input, output, and retained certificate record in an execution. Let $Q_*(n)$ bound the number of charged operations, including failed branches, hierarchy repairs, canonicalization, certificate verification and changes of representation. Fixed constants in local polynomial algorithms must be uniform over the allowed input family. In particular, a handle-type catalog or a frontier alphabet whose size grows with n cannot be hidden inside such a constant.

Proposition 11.1 (Conditional composition bound). *Suppose a sound and complete recognition algorithm has preprocessing time $P(n)$, makes at most $Q_*(n)$ calls, and each call costs at most*

$C(1 + S_*(n))^a$ bit operations for fixed C, a . If, for fixed $\alpha, \beta, \gamma \geq 1$,

$$Q_*(n) \leq \exp(O(\log^\alpha(n+2))),$$

$$S_*(n) \leq \exp(O(\log^\beta(n+2))),$$

$$P(n) \leq \exp(O(\log^\gamma(n+2))),$$

then its total time is $\exp(O(\log^{\max(\alpha, \beta, \gamma)}(n+2)))$. In particular, exponents at most two suffice for $n^{O(\log n)}$.

Proof. The total time is at most $P(n) + CQ_*(n)(1 + S_*(n))^a$. Taking logarithms of the second term adds the bounds for $\log Q_*$ and $a \log(1 + S_*)$. A fixed finite sum of these logarithmic powers is bounded by a constant times the largest one. Adding preprocessing does not change that largest exponent. $\square$

The proposition is an accounting criterion, not a theorem that its hypotheses hold for the present recognizer. Polynomial time for a local query does not bound Q_* or S_* . An inconclusive budget exit does not supply completeness. Nor does a short successful certificate show that a deterministic search finds it within the same bound. All expanded data that must actually be written count towards the relevant size and time.

11.2 Hierarchy depth, restarts and branching are separate parameters

Write H for maximum hierarchy length and G for maximum pattern complexity. In this notation Lackenby's Proposition 9.1 gives the iteration bound $H(G+1)^H$ [2]. Hence a sufficient restriction for this part of the calculation is

$$H \log(G+1) + \log(H+1) = O(\log^2 n).$$

For example, $H = O(\log n)$ and $G = n^{O(1)}$ suffice. If instead $H = O(\log^2 n)$ and $G = n^{O(1)}$, this particular bound gives $\exp(O(\log^3 n))$: quasi-polynomial in the broader sense, but not the specific $n^{O(\log n)}$ target. A polynomial bound on the bit length of G alone does not imply $G = n^{O(1)}$.

The refined construction in Section 10 controls hierarchy length linearly in the initial q -complexity. It does not by itself give logarithmic depth or a quasi-polynomial bound on the restart expression. Conversely, replacing a costly surface-topology query by AHT lowers the cost of one step without lowering the number of lexicographic repairs. These are different improvements and require different proofs.

If a proposed construction has at most $B(n)$ possible children at depth at most $H(n)$, its search tree can contain as many as $\sum_{j=0}^{H(n)} B(n)^j$ nodes. Polynomial branching and logarithmic depth are sufficient for a quasi-polynomial tree-size bound. Polynomial depth is not. Revisiting or modifying a node requires an additional amortized bound, even if the number of distinct nodes is controlled. Memoization helps only after proving that the canonical state retains every relevant boundary marking and that the number of such states is small. A state with polynomially many bits can still have exponentially many possible values.

Falsifiable hierarchy question. For the multisurface and generalized-Heegaard construction, identify a potential that pays both for ordinary progress and for discarding later surfaces after a repair. Can every long execution produce a certified Cheeger-region simplification before the restart cost exceeds a polylogarithmic exponent? Useful negative instances would keep the recorded local complexities small while forcing many successive repairs. Such examples would refute a proposed potential, even if all topology queries themselves were fast. The announced acceleration framework [3] motivates this question; the present kernels do not implement its global argument.

11.3 From normal-coordinate topology to certified cutting

The current normal adapter supplies connectivity and total topological statistics for a supplied surface. A compressed cutting primitive has a larger contract. It must return the complementary handle structure, the parallelity components, the topology and twisting of their base surfaces, and the attachment of their vertical boundaries. These are the kinds of output required by Lackenby's Theorem 14.4 [2]. Matching component counts do not determine these attachments. Even matching unmarked surface types does not determine a map relative to a marked boundary.

The first implementation target is weighted AHT with grouped multiplicity records, followed by a compiler from those records to the required cut incidence data. The weighted orbit theorem is classical [4]; its implementation would extend the current module rather than establish a new existence theorem. Its output must remain grouped when a binary vector describes exponentially many equal components. Explicitly listing one record per component defeats the compression.

A useful contract for testing is a commuting reconstruction diagram: on small inputs, expand the original surface and perform an independent ordinary cut; expand the compressed output; compare the resulting marked cell structures. Include twisted bundles, annular interfaces, self-identifications, boundary-parallel discs and disconnected surfaces. For large inputs, retain locally checkable construction records instead of treating a numerical topology summary as a reconstruction proof.

An equally important target is a verified diagram-to-exterior adapter. It must specify how the peripheral torus corresponds to the knot and how that correspondence changes during simplification. A normal disc whose boundary is essential in a supplied torus certifies compression of that torus. Turning it into a certificate about the input PD diagram requires the exterior correspondence. A complete global theorem also needs a method of finding suitable surfaces within its work budget.

11.4 Frontier and representation bounds

Let w be the number of interface positions in a scanner state, τ the number of retained scalar types, $\mu = \max_v s_v$, E the number of nonzero typed blocks, $\rho = \max_{u,v} \rho_{uv}$, and B the largest packed morphism bit length. The elementary inequalities

$$V \leq \tau \mu^2, \quad P_{\text{span}} \leq E \rho \mu^2$$

describe how these parameters enter the solver. They do not bound the parameters in terms of n . A small coefficient span does not bound multiplicity, and small frontier width does not by itself bound the size of all coefficients or the number of repeated states.

One sufficient approach is to prove a uniform encoding bound of the form $S_* \leq \exp(O(w \log(w+2))) \text{poly}(n)$ and a suitable bound on all visits, then establish $w \log(w+2) = O(\log^2 n)$ for the chosen decomposition or for an exact quotient of its states. This is a proposed route, not a property established for the present scanner. A square-root-width separator bound, even if available for a particular diagram graph, would give a much larger exponent in this model and would not meet the target.

The experiment should record $w, \tau, \mu, E, \rho, B, V, V_F$, reconstruction cost, and the number of repeated states together. Search for families in which one proposed controlling parameter stays fixed while another grows. In particular, a useful challenge to the forest proposal is a family with small V_F but costly dense reconstruction in the original V variables. The current bounded-candidate policy requires that reconstruction, so it cannot be excluded from the accounting.

11.5 A tractable commutant family and its limitations

There is a clean special case in which the forest construction supplies explicit projections, rather than merely a smaller linear system.

Proposition 11.2 (Scalar transports give explicit repeated summands). *Consider an entire finite differential component with no incident differential blocks omitted. Suppose every scalar type has multiplicity s and the selected invertible forest connects all its types. Use the transport matrices P_v from Theorem 7.1. The complete commutant is the common centralizer in $M_s(\mathbb{F}_2)$ of every transported coefficient matrix $P_v^{-1}A_iP_u$.*

If every such matrix is a scalar multiple of I_s , the component is isomorphic to a direct sum of s identical multiplicity-one typed complexes. Explicit compatible matrix units in the original coordinates are

$$E_{ij}^{(v)} = P_v E_{ij} P_v^{-1}.$$

In particular, the s diagonal units are orthogonal rank-one projectors whose sum is the identity. If additional attachment maps are required in the splitting contract, the assertion applies when they also respect these summands; this must be verified, not inferred from the differential.

Proof. There is one root matrix X because the forest is connected. Theorem 7.1 changes every original equation into $X\tilde{A}_i = \tilde{A}_iX$, proving the centralizer statement. Under the scalar hypothesis every $X \in M_s(\mathbb{F}_2)$ satisfies these equations. After changing the basis at type v by P_v , each differential block is a morphism coefficient times I_s , since all of its scalar coefficient matrices are scalar. The s coordinate lines are therefore invariant and carry identical typed differentials. Conjugating the usual matrix units back gives the displayed formula, their multiplication identities, and the rank-one projections. Their images supply the claimed direct sum. The same conclusion for extra attachments follows only if the required invariance or transported block form is checked for those maps as well. $\square$

A sufficient case is that the graph of all nonzero typed blocks is a tree and each block has a one-dimensional coefficient span whose nonzero coefficient matrix is invertible. Choosing those matrices as the forest transports turns every block coefficient matrix into I_s . This is an algebraic family; its prevalence among actual knot-scan components remains an empirical and structural question. The current implementation computes and canonically reconstructs the full commutant. It does not yet bypass the bounded candidate policy with these direct projectors.

Cycles explain a limitation. A non-tree invertible block becomes a holonomy constraint on X , and all noninvertible coefficients impose their own centralizer conditions. If these matrices have only the scalar common centralizer, there are no nontrivial projections in that algebra. Thus a connected invertible forest does not guarantee a useful split. At the other extreme, low-rank coefficient spans need not display an invertible basis element: the two singular matrices $\text{diag}(1, 0)$ and $\text{diag}(0, 1)$ span I_2 . The current forest search can therefore miss an invertible linear combination. Exhausting the span costs 2^ρ candidates, so any wider search needs a stated rank restriction or its own complexity argument.

11.6 Explicit and compressed relator growth

The new shared query costs nearly linearly in the total explicit relator length ℓ , with the stated logarithmic and dictionary-model factors. Its speed does not make ℓ small. Generator substitution can duplicate long words across many relators; a sequence of individually valid moves can therefore create an expensive explicit presentation before the next shortening query is reached.

There is a modest unconditional observation for a restricted phase: if every accepted relator replacement strictly decreases total explicit length, and no intervening operation increases it, the number of such replacements is at most the starting length. This says nothing about whether that phase recognizes every unknot, and it does not bound the length at entry to later phases. A useful next theorem would control both presentation growth and the number of verified moves across the whole producer, including generator eliminations and Whitehead steps.

For straight-line representations, track grammar size, logarithmic represented length, the number of rewritten nonterminals, and the cost of checking each move. A shared fully compressed cyclic index would need to handle donor exclusion and occurrence deletion without expanding the rotations. An incremental explicit index is another concrete target, but reusing an index after mutation requires exact removal of obsolete occurrences. Each proposal should be challenged by presentations where short descriptions expand rapidly or where most indexed occurrences are invalidated by a single certified move.

11.7 Proposed research questions

The following questions are prioritized by how directly they remove a known implementation or proof obstruction. They can be pursued separately, but every claimed global application must satisfy the complete accounting contract above.

- Q1. **Weighted component extraction.** Can the native interval kernel return selected component coordinates and grouped topological weights in polynomial encoded time, including one-sided components and exponentially large multiplicities?
- Q2. **Marked cutting.** What is the smallest attachment record that determines the cut complement and its parallelity bundles relative to all peripheral markings? Can its expansion be independently compared with ordinary cutting on every small test input?
- Q3. **Exterior correspondence.** Can each diagram simplification and exterior construction emit a short composable certificate preserving the peripheral identification needed by the final compressing-disc argument?
- Q4. **Hierarchy repairs.** Is there an amortized potential that bounds the cumulative cost of discarded and rebuilt hierarchy suffixes by $\exp(O(\log^2 n))$, or an explicit family refuting a proposed potential?
- Q5. **Interface compression.** Which exact state equivalence can reduce the effective interface description below the width of the raw diagram decomposition while preserving every future attachment operation?
- Q6. **Natural coefficient ranks.** Are there topologically defined knot families with uniformly small ρ_{uv} but growing ambient coefficient support, and can the rank advantage persist after including all scan and reconstruction costs?
- Q7. **Constructive scalar summands.** Can the transported-projector family of Proposition 11.2 be recognized and split directly under the scanner's full attachment contract, avoiding construction of its entire commutant basis?
- Q8. **Invertible combinations.** For the coefficient spaces actually arising in the scanner, can an invertible linear combination be found with a proved cost smaller than enumerating all 2^p combinations?
- Q9. **Canonical output cost.** Can candidate selection be reformulated in reduced root coordinates with an independently verified equivalence to the existing policy, so that a large full-basis reconstruction is avoided?
- Q10. **Shared compressed word search.** Can capped cyclic windows and distinct-slot donor constraints be indexed in time polynomial in grammar size and logarithmic represented length, without expanding the rotations?
- Q11. **Presentation growth.** For which nontrivial diagram families are both intermediate presentation size and the full verified move count quasi-polynomially bounded, including the substitutions that increase length between shortening phases?

- Q12. Incremental exact indices.** Can relator replacement and deletion update the shared index faster than rebuilding it while deleting every obsolete occurrence and retaining a compact independently checkable match?

11.8 Milestones that would justify stronger conclusions

The next report should distinguish the following outcomes.

1. A completed weighted-orbit and marked-cutting implementation, with independent reconstruction tests and a proved bound on its encoded output.
2. A verified connection from the ordinary knot diagram to the triangulation and peripheral data consumed by the geometric kernels.
3. A structural theorem for a nontrivial diagram family controlling every parameter in Proposition 11.1, rather than only the frontier width or a successful witness size.
4. A constructive implementation of Proposition 11.2, checked against the maintained attachment contract and measured on actual differential components.
5. A complete presentation-growth and move-count bound for a specified group-simplification family, or an exact compressed replacement for the steps that defeat such a bound.
6. A hierarchy theorem paying simultaneously for depth, branching, repairs and representation changes, with every local oracle instantiated.

Any one of these would close a clearly identified gap. A general $n^{O(\log n)}$ conclusion requires enough of them, or an alternative complete route, to meet a uniform total-work bound on every input.

A Reproducing the numerical report

From the repository root, first verify and apply `integration.patch` against the pinned baseline. Detailed commands and a checksum manifest are included in the archive's README. The ordinary full-suite command is run from `Topology/UnknotRecognition/fast`:

```
PYTHONPATH=. python3 -B -m unittest discover -s tests -v
```

The maintained suite has sibling reference dependencies in `reports/24/reference`, `reports/26/detshadow` and `reports/28/src/closure_reset`. They must be present at the pinned commit. A sparse checkout of `fast/` alone is insufficient for this full gate. The independent Regina comparisons are optional tests, but they were available and ran in the reported environment.

The four experiment entry points, run from `fast/`, are:

```
python3 -B interval_research/benchmark.py \
--output interval_research/results.json
python3 -B normal_orbit_research/benchmark.py \
--output normal_orbit_research/results.json
python3 -B benchmark_coefficient_span.py \
--output coefficient_research/results.json
python3 -B cyclic_overlap_research/benchmark.py \
--output cyclic_overlap_research/results.json
python3 -B cyclic_overlap_research/make_summary.py
```

Use new output paths to retain the shipped samples unchanged. The benchmark scripts record exact answers and reject disagreements before producing a successful report. The interval, coefficient and relator benchmarks also check that their measured source files did not change during the run. The normal series records end-of-run source hashes; the final static audit checks all recorded hashes against the supplied files. Random seeds, warmup policy, repeat counts and source digests are stored in the JSON.

To regenerate the figures and tables, run `render_results.py` from the article directory; it reads the four archived result files at their repository-relative locations. Matplotlib is required only for this step. The supplied figure PDFs and tables allow the article to be built without rerunning any experiment:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

B The complete whole-query corpus

Table 9 provides the per-input full-query medians in addition to the focused Gordian comparison. Inputs that never reach the optimized query are necessary controls. Their nearly identical timings should not be interpreted as evidence for, or against, the local index theorem. Every measured run and excluded warmup is retained in the raw corpus.

Case	n	E pair ms	E adapt ms	E ratio	C pair ms	C adapt ms	C ratio
survivor-00	16	3.153	3.186	0.954	11.808	12.613	0.929
survivor-01	15	2.398	2.384	1.033	10.594	9.966	1.082
survivor-02	15	2.866	2.940	0.960	13.820	13.876	0.942
survivor-03	13	2.168	2.406	0.969	8.431	9.904	0.906
mirror-03	13	4.945	4.580	1.057	16.982	16.328	0.999
survivor-04	13	1.774	1.694	1.045	7.898	7.900	0.996
survivor-05	13	3.078	3.115	0.971	13.460	12.339	1.081
survivor-06	12	2.288	2.127	1.035	6.986	6.999	1.006
survivor-07	12	3.058	3.063	1.010	13.952	15.293	0.838
survivor-08	11	1.318	1.414	0.992	5.224	5.567	0.941
mirror-08	11	4.584	3.933	1.126	13.204	12.957	1.010
survivor-09	11	1.512	1.436	1.053	5.317	5.305	0.977
survivor-10	11	1.313	1.404	0.945	4.759	5.039	0.964
survivor-11	11	1.534	1.716	0.928	6.225	6.378	0.997
gordian	141	830.706	490.701	1.766	1357.319	1045.271	1.297
trefoil	3	0.030	0.031	0.996	0.029	0.028	1.009
figure_eight	4	0.032	0.032	1.016	0.030	0.030	0.988
conway	11	0.609	0.593	1.085	0.659	0.669	1.028
kinoshita_terasaka	11	0.626	0.664	0.930	0.627	0.653	0.960

Table 9: All whole-query cases. E and C denote the explicit and compressed configurations. All statuses are complete and agree with the listed corpus expectations in the raw data.

C Attribution and artifact provenance

The maintained pipeline and layered-torus fixtures belong to the ProveIt baseline [1]. The native orbit implementation was written independently from the AHT paper; no third-party GPL orbit implementation was imported or copied. The stronger threshold uses Fine–Wilf periodicity [5]; the graph interpretation has a substantial existing literature, including Rankin [6]. The normal connectivity reduction and weighted-orbit existence theorem are classical AHT results. Suffix automata are classical [7], and the coefficient and forest identities are elementary linear algebra applied to the maintained typed solver. Bar-Natan’s local and formal-complex methods provide the relevant Khovanov computational setting [8, 9].

The new package claims these particular implementations, integration decisions, proofs of their contracts and reproducible observations. It does not claim priority for the general mathematical ingredients. The normal validator is adapted from report 36’s native certificate checker; the inherited fixture file is identified separately in the provenance record. Regina is used only as an external independent test oracle and is not redistributed as production source in this archive.

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